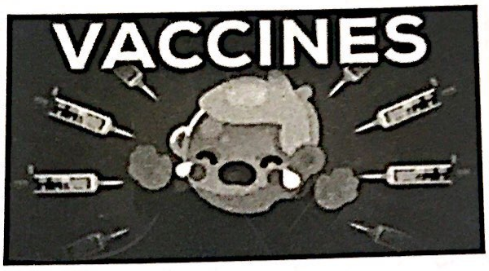
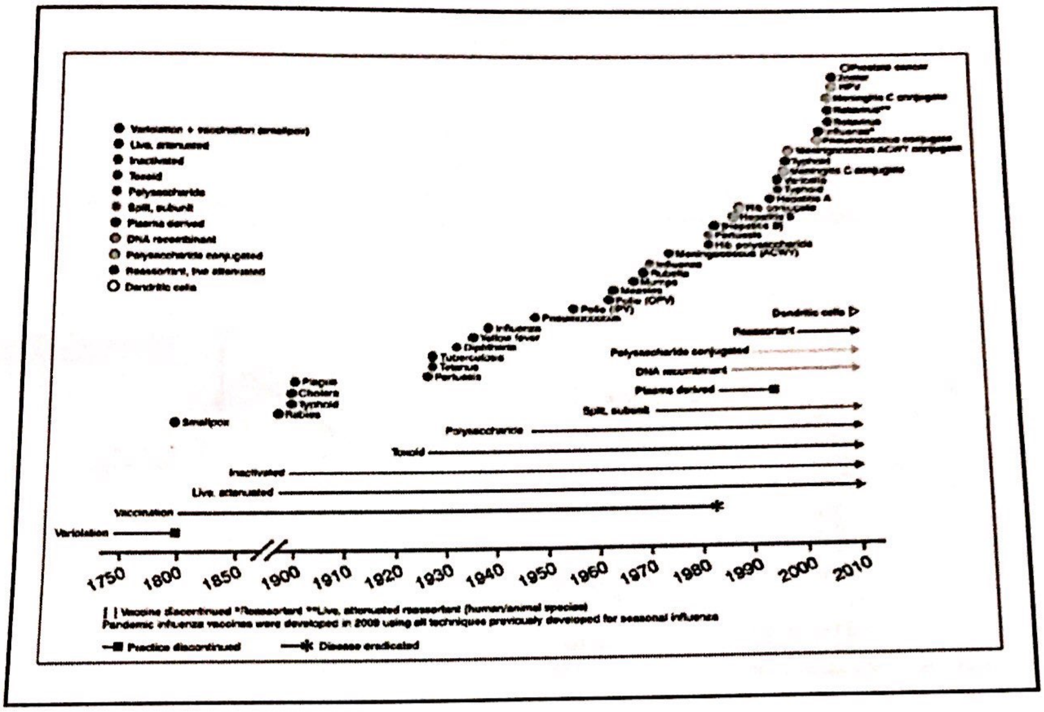
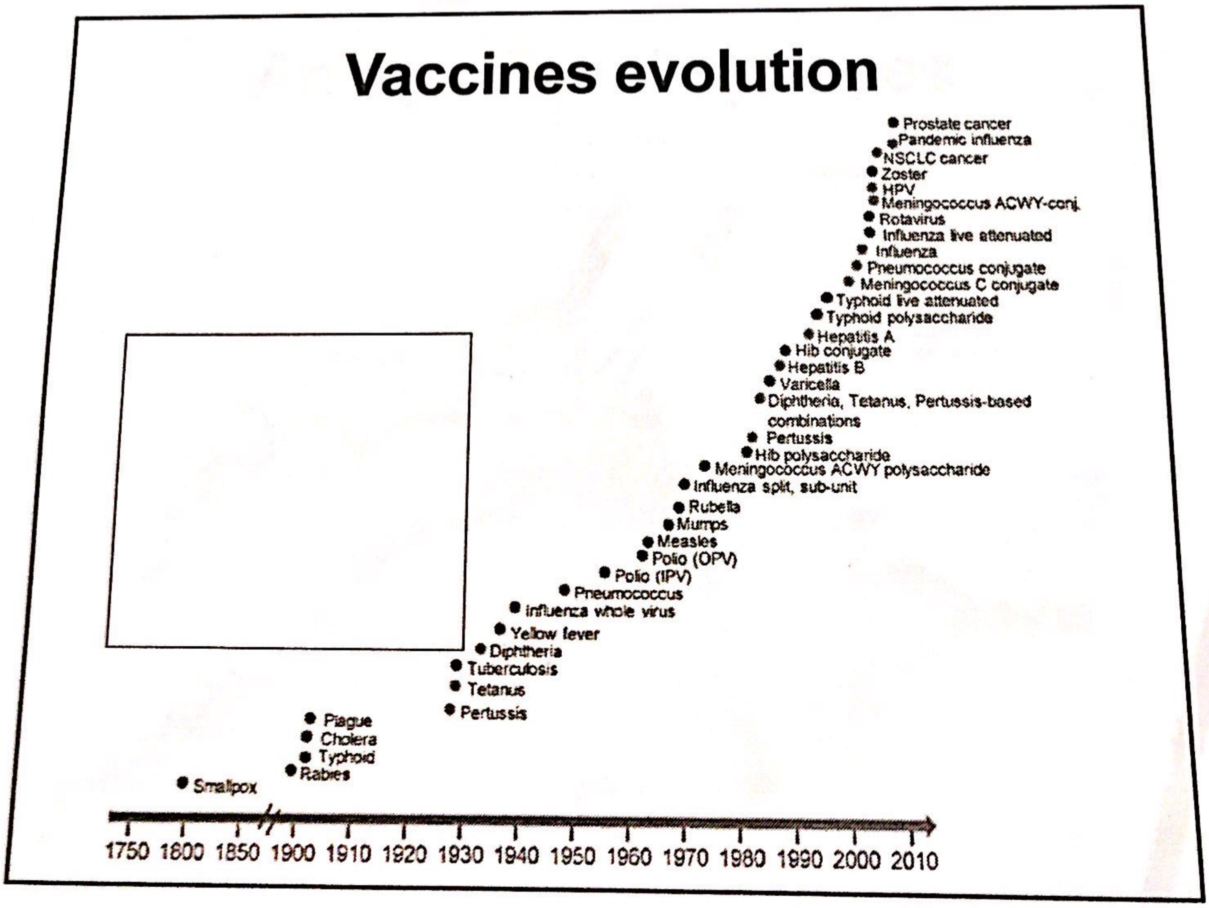
Vaccines

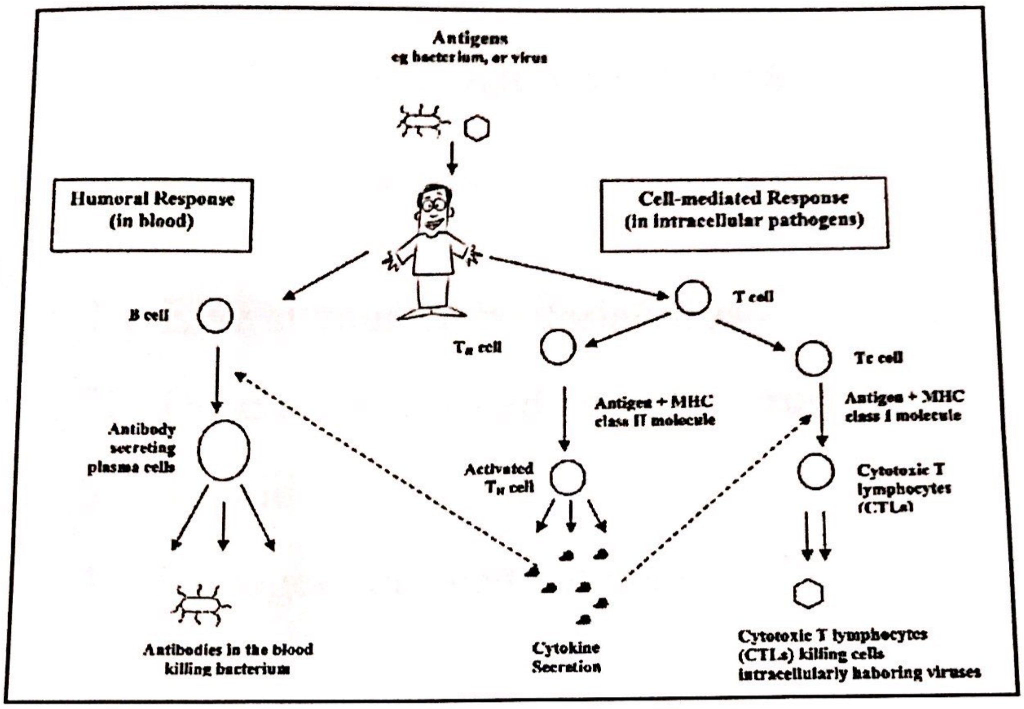


Vaccines

- Vaccines are preparations of dead or weakened pathogens, or their products, that when introduced into the body, stimulate the production of protective antibodies or T cells without causing the disease
- The name ‘vaccine’ comes from the Latin vacca (for cow). The earliest vaccination was done using the cow pox virus (which causes the disease in cow) as a vaccine against small pox in humans.
- The English physician, Edward Jenner carried out the above vaccination in the late 18th century .

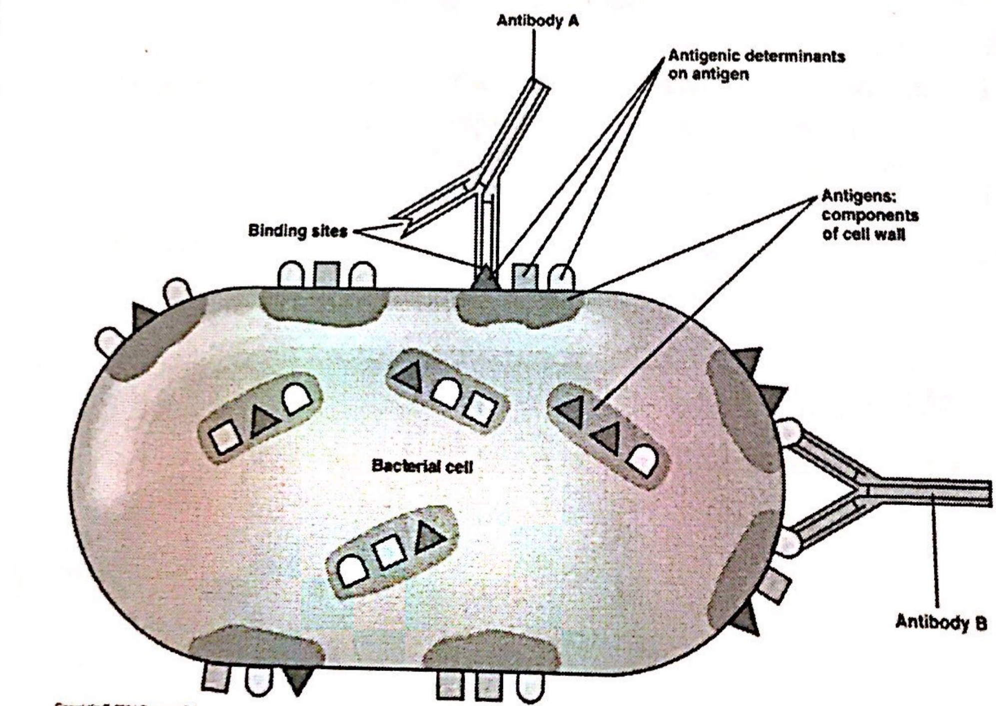

Variolation + vaccination (smallpox)
Live, attenuated
Inactivated
Toxoid
Polysaccharide
Split, subunit
Plasma derived
DNA recombinant
Polysaccharide conjugated
Reassortant, live attenuated
Dendritic cells
Smallpox
Plague
Cholera
Typhoid
Rabies
Pertussis
Tetanus
Tuberculosis
Diphtheria
Yellow fever
Influenza
Polio (IPV)
Polio (OPV)
Measles
Mumps
Rubella
Influenza
Hib polysaccharide
Pertussis
Hepatitis B
Hib conjugate
Hepatitis A
Typhoid
Varicella
Typhoid
Pneumococcus conjugate
Rotavirus
HPV
Zoster
Prostate cancer
Dendritic cells
Reassortant
Polysaccharide conjugated
DNA recombinant
Plasma derived
Split, subunit
Polysaccharide
Toxoid
Inactivated
Live, attenuated
Vaccination
Variolation
1750 1800 1850 1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000 2010
Pandemic influenza vaccines were developed in 2009 using all techniques previously developed for seasonal influenza
Practice discontinued
Disease eradicated


Vaccines evolution
Prostate cancer
Pandemic influenza
NSCLC cancer
Zoster
HPV
Meningococcus ACWY-conj.
Rotavirus
Influenza live attenuated
Influenza
Pneumococcus conjugate
Meningococcus C conjugate
Typhoid live attenuated
Typhoid polysaccharide
Hepatitis A
Hib conjugate
Hepatitis B
Varicella
Diphtheria, Tetanus, Pertussis-based combinations
Pertussis
Hib polysaccharide
Meningococcus ACWY polysaccharide
Influenza split, sub-unit
Rubella
Mumps
Measles
Polio (OPV)
Polio (IPV)
Pneumococcus
Influenza whole virus
Yellow fever
Diphtheria
Tuberculosis
Tetanus
Pertussis
Plague
Cholera
Typhoid
Rabies
Smallpox
1750 1800 1850 1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000 2010



* properteis of a good antigen:
high MW, complex chemicals (preferly proteins), forgein

Antigens and epitopes



Traditional vaccines

Traditional vaccines can be classified into four broad groups:

1. Live attenuated bacteria/virus
2. Dead or inactivated bacteria/virus
3. Toxoids
4. Pathogen- derived antigens

Traditional vaccines

Live attenuated

- Attenuation is the process of weakening or reducing the virulence of a microorganism by using empirical procedures like prolonged storage and cultivation under suboptimal conditions (passaging).

(applying stress) (decreased pathogenicity)

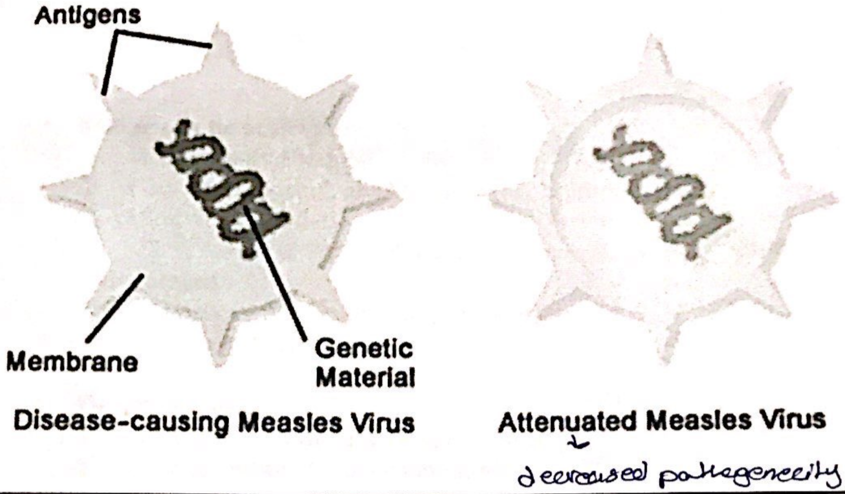
$\rightarrow$ A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ D

* in the same animals

* passed 30-40 H

- Viruses are also attenuated in appropriate animal cell culture systems for many generations under a stressful environment
- **Advantages:**
  1. Low cost of preparation
  2. The ability to elicit the desired immunological response in a single-dose application.
- **Disadvantages:**
  1. The potential to revert and become virulent
  2. Limited shelf life.

Traditional vaccines
Live attenuated



Traditional vaccines
Live attenuated

Table 11.1 Cell culture systems used for passaging viral particles for use as vaccines

Cell culture system	Viral particle/vaccine
Chick egg embryos	Yellow fever virus
Chick egg embryo cells	Measles virus, mumps virus
Monkey kidney tissue culture	Polio (Sabin's vaccine)
Human diploid fibroblasts	Hepatitis A viral vaccine

fertilized egg

Traditional vaccines
Dead, Inactivated Vaccines

- Consist of suspensions of fully virulent organisms killed as mildly as possible in order not to destroy the antigenic determinants on the organism.

→ bc we want to kill the bacteria but keep the surface antigens alive (autoclaving will damage them)

- **Killing can be achieved:**
  1. Heat (usually about 60° C for 1 hour)
  2. Chemicals (phenol, alcohol, formalin, b-propiolactone)
  3. Ultraviolet irradiation

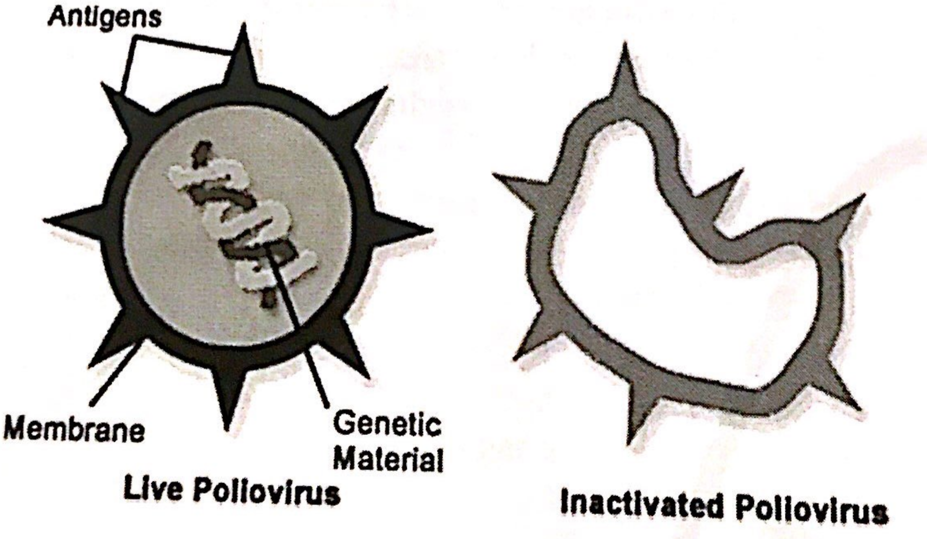
↳ (formaldehyde, is the most used)

↳ damage the DNA of the bacteria but not the surface antigen

- **Advantages:**
  1. No chances of reversal into a virulent strain
  2. Relatively stable shelf life.

- **Disadvantages:**
  1. The higher cost of vaccine development
  2. The possibility of reduced immune

Traditional vaccines
Dead, Inactivated Vaccines



Traditional vaccines
Toxoids

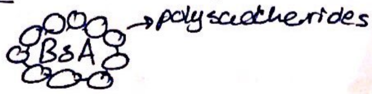
- A toxoid is derived from the treatment of active toxin produced by the bacterium with formaldehyde
- Two commonly toxoid-based vaccine preparations are tetanus and diphtheria vaccines.
- Tetanus toxin is produced by *Clostridium tetani* and diphtheria toxin from *Corynebacterium diphtheriae* which is also used as a toxoid for the treatment of whooping cough.

*toxins are proteins
*toxoids are inactive toxins

drophobi
drophib

Traditional vaccines
Pathogen-Derived Antigens

- The pathogen-derived antigens comprise of antigenic portions of the pathogens which are generally surface derived, most commonly surface polysaccharides
- The antigenicity of surface polysaccharides can be improved by conjugating them with a protein-based antigen.

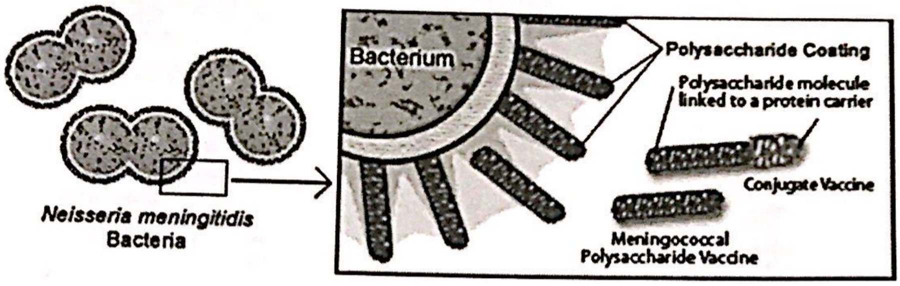


- *Haemophilus influenzae* capsular polysaccharide has been conjugated with diphtheria toxoid or the outer membrane protein of *Neisseria meningitides.*

how to isolate membrane protein?
break bacteria, centerfugation

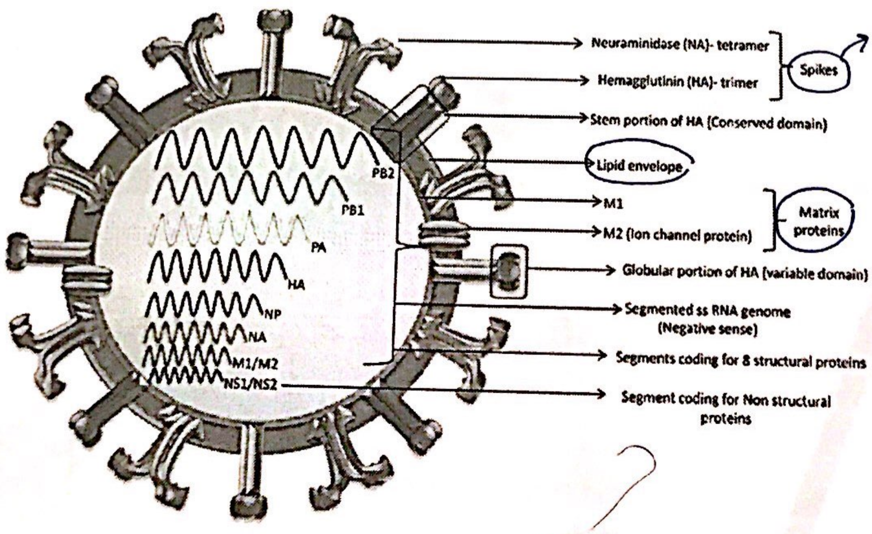
*polysaccharides are not good vaccines because they're not chemically complexed enough

Traditional vaccines
Pathogen-Derived Antigens



*the purpose of vaccines is provocing the immune system to produce memory antibodies

A comprehensive review of antigens on the equine influenza virus



only for viruses, the purtruding antigens

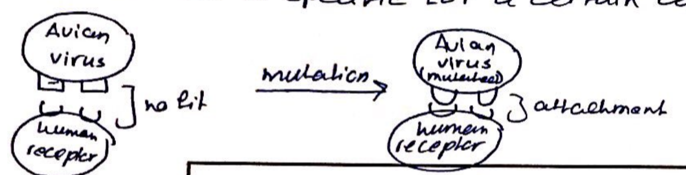
*candidates in the production of vaccines

Table 11.2 Representative examples of traditional vaccine preparations which are being used clinically الدواعي

Product	Vaccine class	Description	Application
BCG (Bacillus Calmette–Guerin) vaccine	Live attenuated vaccine	Attenuated strain of *Mycobacterium tuberculosis*	Immunisation against tuberculosis
Cytomegalovirus vaccine	Live attenuated vaccine	Attenuated strain of human cytomegalovirus	Immunisation against cytomegalovirus
Measles vaccine	Live attenuated vaccine	Attenuated strain of measles virus	Immunisation against measles
Poliomyelitis vaccine	Live attenuated vaccine	Attenuated strains of poliomyelitis virus	Immunisation against polio
Rabies vaccine	Inactivated vaccine	Inactivated rabies vaccine	Immunisation against rabies
Japanese encephalitis vaccine	Inactivated vaccine	Inactivated Japanese encephalitis virus	Immunisation against Japanese encephalitis
Pertussis vaccine	Killed vaccine	Killed strain of *Bordetella pertussis*	Immunisation against whooping cough
Hepatitis A vaccine	Inactivated hepatitis A virus	Formaldehyde-treated hepatitis A virus	Prevention against hepatitis A
Diphtheria vaccine	Toxoid	Treatment of diphtheria toxin by formaldehyde	Prevention of diphtheria by immunisation
Tetanus vaccine	Toxoid	*Clostridium tetani* toxin treated with formaldehyde	Prevention of tetanus by immunisation
Pneumococcal vaccine	Pathogen-derived antigen	Mixture of purified surface polysaccharide antigens obtained from different serotypes of *Streptococcus pneumoniae*	Immunisation against infections caused by *Streptococcus pneumoniae*
Hepatitis B vaccine	Pathogen-derived antigen	Suspension of surface antigen of hepatitis B (HBsAg)	Immunisation against hepatitis B
Meningococcal vaccine	Pathogen-derived antigen	Purified surface polysaccharide antigens of one or more strains of *Neisseria meningitidis*	Immunisation against infections caused by *Neisseria meningitidis*

drophobi
drophibic

* how the virus is specific for a certain cells? (lock and key)



this explains how animal viruses could infect humans

Modern vaccines

The modern vaccines can be classified as

1. Subunit (part of the cell not the whole)
2. DNA-based recombinant (genetic vaccines) (genes are targeted)
3. Peptide vaccines (synthesize peptides and mimik the surface proteins)

* the part that the antibody recognizes is an epitope
↓
(4-5 amino acids)

Modern vaccines
Subunit vaccines

- Subunit vaccines comprise of a part of a bacteria or virus which can best stimulate the immune systems without rendering the person susceptible to the disease.
- The subunits generally comprise of envelope proteins and antigenic surface proteins.
- There are two methods of developing a subunit vaccine:
  1. To grow the bacteria in the laboratory and then break apart to recover and purify the desired antigens
  2. By using recombinant DNA technology clone and express the antigenic moiety in a non-pathogen, generally referred to as recombinant subunit vaccine.

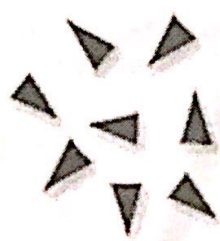
→ envelope proteins are recognize by the immune system when the antigen is shopped

rophobi
drophibi

* PG120, PG140, PG160 are glycoproteins that make the spikes
* a good subunit vaccine should be:
very pathogenic, provoke the immune system in a good way

Modern vaccines
Subunit vaccines

Live Hepatitis B Virus



Hepatitis B Subunits

Modern vaccines

examples of

Subunit vaccines

Table 11.3 Clinically used recombinant subunit vaccines

Product	Description	Company	Application
Recombivax	rHBsAg produced in *Saccharomyces cerevisiae*	Merck	Prevention of hepatitis B
Engerix-B	rHBsAg produced in *Saccharomyces cerevisiae*	GSK	Prevention of hepatitis B
Hepacare	rS, pre-S and pre-S2 hepatitis B surface antigens produced in murine cell line	Medeva Pharma	Hepatitis B immunisation
Twinrix	Recombinant protein of hepatitis B + hepatitis A killed virus	GSK	Hepatitis immunisation
HBVAXPRO	rHBsAg produced in *Saccharomyces cerevisiae*	Aventis Pharma	Hepatitis B immunisation
Gardasil	Recombinant VLPs assembled from the L1 proteins of HPV types 6, 11, 16 and 18	Merck	Immunisation against human papillomavirus

cell line in mice

prevent curvix cancer up to 90%.

papillomavirus can cause curvix cancer, transmitted sexually

* there's a strong association between HPV types 6, 11, 16, 18 and curvix cancer, even a bigger association than smoking with lung cancer

Modern vaccines

Subunit vaccines

Virus-Like Particles (VLPs)

- Virus-like particles are basically viral structural proteins that are expressed in cells and possess native conformational epitope.
- **VLPs** do not contain the genome and therefore cannot spread infection.
- **VLPs** have the capacity to spread both cellular and humoral responses. Both Recombivax and Engerix-B were first to use virus-like particles.

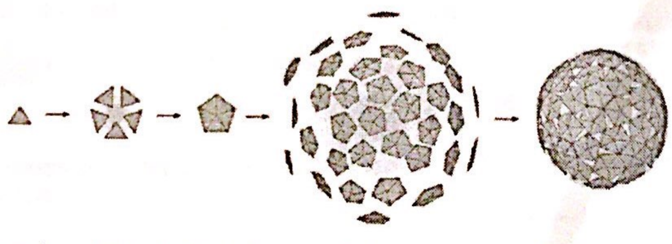


Fig 1. Schematic picture of formation of L1 proteins into HPV VLP.

Modern vaccines
Subunit vaccines

Virus-Like Particles (VLPs)

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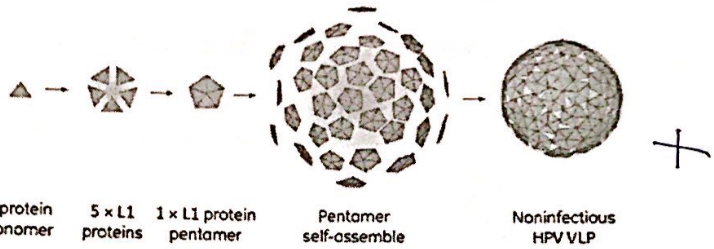
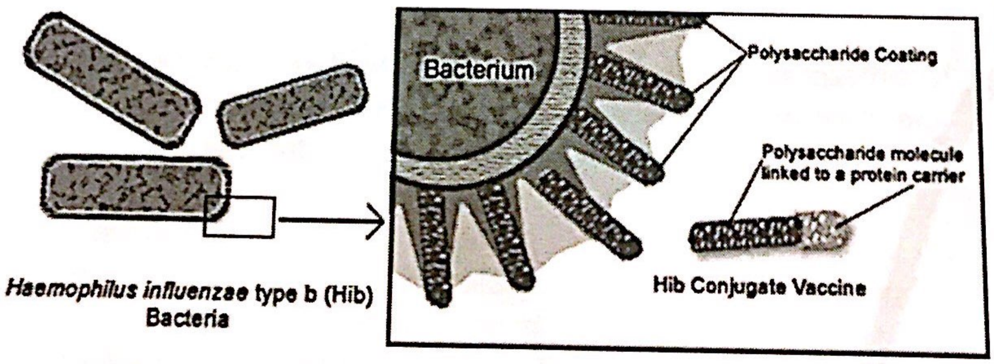


Fig 1. Schematic picture of formation of L1 proteins into HPV VLP.

Modern vaccines
Subunit vaccines

Conjugate Vaccines



١٤٤١

Recombinant vaccines

The recombinant vaccines basically comprise of:

1. Live genetically modified organisms
   (genetically modified to be non toxic)
2. Recombinant subunit vaccines
   (Genetic)

~~3. Genetic vaccines!~~

habity
hibic)

Recombinant vaccines

Live Genetically Modified Organisms

- Bacteria or viruses with one or more deleted or inactivated genes or else could be vaccines carrying a gene from other disease agents which are referred to as vaccine vectors.

- Two (double knockout) or more genes are deleted or inactivated so that the vaccine remains stable and does not revert back into a pathogenic state.

- This type of vaccine development requires knowledge about the gene(s) responsible for the pathogenicity of the organism and assumes that those genes are not the same genes governing viability and the ability of the modified organism to induce an immune response.

* you can't knockout housekeeping genes

* في بعض الحالات بستعملو
امراضا ثانية عشان نعمل
immunity لامراضنا اخطر
مثلا بحقنو بالانسان جدري
البقر لانه اعراضه اخف و
بنفس الوقت الجسم بعمل
immunity لجدري الانسان
لانه الفيروسين بتشابهو
بال surface proteins

Recombinant vaccines

Genetic or DNA Vaccines

- The recombinant vaccines only comprise of the DNA.
- DNA vaccines consist of circular pieces of DNA called the plasmids which contain a foreign gene from the disease- causing agent and a promoter which is helpful in expressing that gene as a protein in the target animal
- Bird flu DNA vaccine was approved in June 2006, and subsequently a veterinary vaccine for horses to prevent the West Nile virus was approved in 2007.
- The use of DNA vaccines for humans is in various stages of development for targets such as HIV, cancer and multiple sclerosis

*DNA vaccines are safe

*immunize
↓ within hours
antibodies
↓ 1week-10 days
removal of antibodies and memory cells stay

~~*proteins are the most~~
*protenies antigens are the most immunogenic bc proteins are more complex
*when we immunize we're not interested in the antibodies that are produced right away, we're interested in the memory cells

1

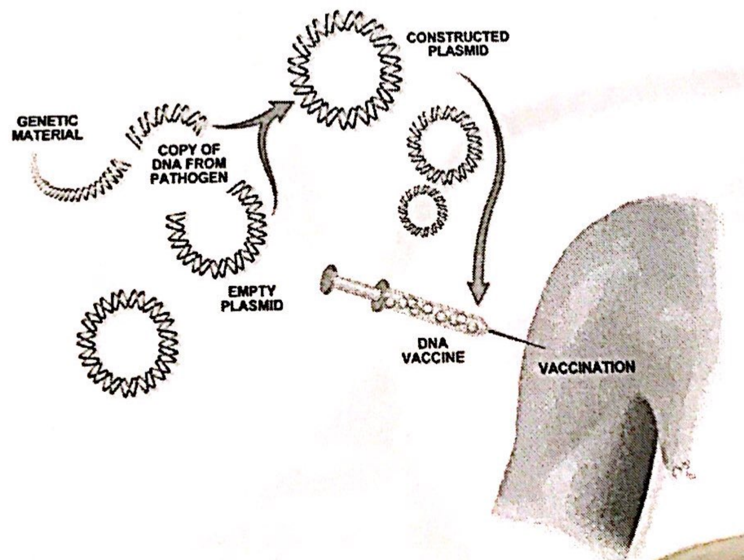
*plasmids : small circular DNA
despinsable يمكن الاستغناء عنه

← في سلايد ٨ مهم

*electroporation: the method used to introduce plasmids by exposing the bacteria to an electric current which will open pores so that the plasmid would come in (the plasmid should be with a strong promoter)

Recombinant vaccines

Genetic or DNA Vaccines



the making of a DNA vaccine:

① the RNA (single stranded genetic information) is extracted from the west Nile virus (WNV) (destroying the virus).
RNA is converted to DNA (double stranded genetic information)
The DNA represents the WNV genome

② the genetic sequence for WNV is generated from the DNA

③ based on the DNA sequence, primers (short sequences of DNA) specific to the Prm and E gene regions are produced. These primers will in turn be used to generate a cDNA fragment containing both the Prm and E genes

④ the cDNA fragment is then inserted into a circular piece of DNA called a plasmid

⑤ the plasmid carrying the Prm and E genes are grown in large quantities in bacteria and purified by column chromatography

⑥ the purified DNA plasmids carrying the Prm and E genes make up the invistigational vaccine.

* Prm gene in codes for the WNV transmembrane protein M
E " " " " " " " E

* once in the body, the DNA plasmid vaccine direct the cells to manifacture the M and E proteins, the immune system should respond by making a defence against the M and E proteins that would protect an individual from a natural ~~#~~ WNV infection

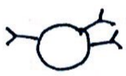
* we have B cell memory and T cell memory

Recombinant vaccines

Peptide Vaccines

→ not recombinant

- Chemically synthesized peptides comprising of 8–24 amino acids. epitopes : (4-5 ~~antigens~~ amino acids)
- As compared to proteins, these are relatively small and have been referred to as peptidomimetic vaccines as they mimic the epitopes.
- Complex structures of cyclic components, branched chains or other configurations can be built into the peptide chain, and thus with these kind of conformations, they mimic the epitopes
- A critical aspect of peptide vaccines → they are too small and must be conjugated to be able to use it as a vaccine is to produce 3D structures similar to the native epitopes of the pathogen. (producing an epitope that'll fit like lock and key with the antibody)


→ epitopos

*why is there no actual treatment for HIV ?

because it's an RNA virus that keeps changing in shape because there's no proofreading thus there are so many mutations